

The Accidental Discoveries of Everyday Objects

Necessity may be the mother of invention, but it seems that quirkiness may have its part to play as well. Many of the greatest discoveries of our times have surprising stories behind them. It seems like many people have achieved greatness purely on accident. Studying these seem to point towards what makes inventors tick, and what the art of invention itself is.

Probably the most familiar story of this kind is Alexander Fleming and the invention of penicillin. In 1928, Fleming went on vacation mid-way through an experiment. In a careless (or rather, messy) moment, Fleming left a dirty petri dish behind. He returned to find the bacteria culture spread all over the plate. The bacteria culture in question was *Staphylococcus* – a bacterium that caused boils, sore throats and abscesses. However, Fleming quickly noticed an anomaly. The petri dish was dotted with colonies of the bacterium, except for one area where there happened to be a mould growing. Fleming extrapolated this to mean that the mould contributed to the killing the bacteria. He was right, of course – the mould was a rare strain of *Penicillium notatum*, and secreted something that stopped bacterial growth.

Next, we have X-Rays. Literally named X-Rays to hint at the fact that their origin was unclear at that time, it was German physicist Wilhelm Röntgen who discovered X-rays in 1895. Röntgen noticed in between his experiments with cathode ray tubes that there was an odd glow somewhat near the tube. These rays were rightly identified by Röntgen as being a new type of radiation, giving them their “X”. As we know, by now, X-Rays are high frequency rays that have enough energy to penetrate through, amongst other things. Interestingly enough, Röntgen also became the first man to take a medical X-Ray with him taking a picture of his wife’s hand, and scaring the woman in the process.

Without anaesthesia, your trips to the dentist would probably be worse than they are now. There is some contention about who exactly discovered anaesthesia, but its origins were due to the popular use of ether and nitrous oxide (laughing gas) in partying. “Laughing parties” as they were called, involved a lot of people howling and laughing and having a good time. Eventually someone (most cite Horace Wells, also a dentist) noticed that these laughing parties also seemed to be oblivious to pain, leading to nitrous oxide being an early form of anaesthesia.

Finally, for a more explosive discovery, we look towards John Walker, an English chemist, who in 1826, became the first to discover a friction match, a stick coated with a mixture of chemicals that would light up when rubbed against phosphorous lined paper. It was a discovery born out of frustration. Walker, in 1826, noticed a dry lump at the end of his mixing stick. Trying to get rid of the lump, he struck the stick against those chemicals, which happened to include phosphorous compounds. Flames erupted and an invention was made.

Percy Spencer was a Raytheon engineer testing magnetron - a source of vibrating electromagnetic waves of a certain frequency. They were used in military level equipment at the time, and Spencer was trying to improve their power level. Spencer was no stranger to inventing, however, the man had earned several patents while working on this problem. So, it is no surprise that when working with magnetron led to a chocolate bar in the man’s pocket melting, he immediately recognised it as something remarkable. In keeping with the scientific method, Spencer repeated the experiment – first with egg, which didn’t go so well, and then with corn, which went wonderfully. Thus, the microwave was born with popcorn accompanying it.

It seems so odd that ubiquitous objects like matches and anaesthesia were discovered in such surprising ways. What seems even stranger is how these were not planned at all – they were a matter of random coincidence (and some hard work). In the end, it seems that the common thread

in these accidental discoveries is the fact that the people who made those mistakes cared enough to re-examine them. Maybe that is the lesson you need to learn to adopt to your life. Mistakes don't define us until we define them, and the only way to do that is to study the accident carefully. There's a lot to learn in accidents, and it seems the world itself wouldn't be the same without them.